Denis Bakin

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About me

Computer vision engineer with 2 years of experience in a robotics lab: stereo camera calibration, real-time detection and segmentation inference, ROS2 and C++.

Skills

ML/CV: Object detection and segmentation, stereo vision, camera calibration, inference optimization

Technologies: Python, C++, PyTorch, OpenCV, ROS2, Docker, Git, Linux, GitHub Actions

Languages: Russian (native), English (C1)

Education

Higher School of Economics, Applied Mathematics and Informatics — Sept 2022 – Present

- Specialization: Machine Learning and Applications

Professional Experience

IITP RAS, Color Research Lab, Student — Sept 2025 – Present

- Bachelor's thesis: "Weakly supervised semantic segmentation for crops production"
- Built a full WSSS pipeline for the PlantSeg dataset (115 disease classes, 35 plant species): MCTformer-V2, DenseCRF, AffinityNet, Random Walk, WeakCLIP
- Investigating applicability of foundation models (SAM, WeakCLIP) vs. classical WSSS methods to narrow domains with limited annotation

HSE Robotics Laboratory, Research Intern — July 2023 – May 2025

- Developed a stereo calibration pipeline: computing intrinsics and extrinsics for a camera pair
- Implemented a frame capture and synchronization pipeline for high-speed cameras via ROS2 (C++)
- Built a detection and segmentation inference pipeline achieving high FPS (TorchScript)
- Set up CI with GitHub Actions for lab projects

Projects

Handy — real-time table tennis ball tracking — GitHub

- Team project at HSE Robotics Laboratory. Technologies: Python, C++, OpenCV, PyTorch, ROS2
- Developed a calibration and detection visualization tool (official and full demo)

TouchAndGo — navigation for visually impaired via stereo vision — GitHub

- Depth map generation from a stereo camera pair to simulate 2D lidar and control vibration motors
- Technologies: Python, OpenCV, Raspberry Pi

Other

HSE Lyceum, C++ and Algorithms Teacher. Developing curricula, problem sets, and entrance exams — 2024 – Present

Private Tutor in mathematics, computer science, and programming — 2022 – Present